



Vũ Minh Trí

Position Applied: AI Engineering
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OBJECTIVE

As an IT student passionate about coding, artificial intelligence, and innovation, I am seeking an **AI Engineering** internship to improve my skills through real-world projects. I enjoy learning new technologies, tackling complex challenges, and **applying AI to enhance productivity and problem-solving.**

EDUCATION

2022 - 2026 **Hanoi University of Science and Technology**
Major: Vietnam–Japan Information Technology (Hedspi)

SKILLS

Programming Languages	JavaScript, Java, Python, C, C++
Deep Learning	CNN
Machine Learning	Machine Learning Algorithms (Linear Regression, SVM, Decision Trees...)
Python- library	NumPy, pandas, scikit-learn, matplotlib,TensorFlow
Database	PostgresSQL, SQLite
DevOps & Tools	Git (GitHub), Docker, Postman

PROJECTS && WORK EXPERIENCE

AI Resume Matching

Company: VnextSoftware

Position: Intern NodeJS

Description: Support and participate in development a system that uses the Google Gemini to analyze candidate resumes and job descriptions, compute similarity, and rank candidates based on relevance to support the recruitment screening process.

Technologies: TypeScript · Node.js · NestJS · PostgreSQL · TypeORM · pgvector · Docker · JWT · Bcrypt · class-validator · zod · Tesseract · pdf-parse · Mammoth · Sharp · Pino · Axios

Team size: 12 members

Responsibilities

- **TypeScript, Node.js, NestJS** – built scalable backend APIs and structured application architecture
- **PostgreSQL, TypeORM, pgvector** – managed relational data and vector storage for semantic search (RAG)
- **Docker & Docker Compose** – containerized services and handled environment setup/deployment
- **JWT Authentication, Bcrypt** – implemented authentication and secure password hashing

July 2025 - Dec 2025

- **Validation (class-validator, zod)** – ensured data integrity and request validation
- **Document Processing & OCR (Tesseract, pdf-parse, Mammoth, Sharp)** – extracted and processed data from PDFs, images, and DOCX files
- **Logging & HTTP (Pino, Axios)** – handled structured logging and external API communication
- **Testing (Jest, Supertest)** – wrote unit and integration tests for backend services
- **Tooling (ESLint, Prettier)** – maintained code quality and consistent formatting

RAG - Document-Based QA Chatbot

Feb 2026 - Now

Description: A Document-based Question Answering (QA) Chatbot is a system that answers user questions by using information stored in documents such as PDFs, Word files, or text data. It is designed to help users quickly find useful information without reading the entire document.

Team size: 1 member

Code: **Link github:** <https://github.com/akari-zero-1/RAG---beginning>

Technologies: Python • Jupyter Notebook • LangChain • Groq (Llama 3.x) • Hugging Face Embeddings (all-MiniLM-L6-v2) • ChromaDB • RAG (Hybrid/BM25, Multi-Query, RRF, Cohere Rerank, History-aware) • Unstructured • Tesseract OCR

- Responsibilities**
- Designed and implemented an end-to-end **RAG pipeline** (ingestion → chunking → embedding → indexing → retrieval → answer generation)
 - Applied advanced retrieval strategies including **Hybrid (BM25 + dense)**, **Multi-Query**, **Reciprocal Rank Fusion (RRF)**, **history-aware retrieval**, and **reranking**
 - **Integrated LLM, embeddings, and vector database** (Groq Llama, Hugging Face, ChromaDB)
 - Processed **multi-modal** documents (text/PDF) using Unstructured and OCR (Tesseract)

CNN For Medical Image Processing

Jan 2025 - Feb 2025

Description: Developed a deep learning model using Convolutional Neural Networks (CNN) to classify chest X-ray images into NORMAL and PNEUMONIA.

Team size: 1 member

Link github: <https://github.com/akari-zero-1/AI-introduce>

Technologies: Python • TensorFlow • Keras • CNN • NumPy • Matplotlib • ImageDataGenerator

- Responsibilities**
- Performed **data exploration** and **preprocessing** for chest X-ray image dataset
 - Implemented a **Convolutional Neural Network (CNN)** model to classify images into NORMAL and PNEUMONIA
 - Applied image preprocessing and data augmentation using **ImageDataGenerator**
 - Trained and validated the model using **TensorFlow/Keras**
 - Evaluated model performance using **accuracy, confusion matrix**, and classification report
 - **Visualized** training results with loss and accuracy curves
 - **Implemented prediction** function to classify new X-ray images